

Bioinformatics session

4th annual workshop on
bioinformatics and variant
interpretation in InPreD

https://inpred.github.io/26-09_bioinfo_ws/



2. Containerization



A brief history

1979	<code>chroot</code> system call (changing the root directory of a process and its children to a new location in the filesystem) in Uni V7 considered as beginning of process isolation
2000	FreeBSD Jails allows administrators to partition FreeBSD computer system into several independent, smaller systems (<i>jails</i>) – with the ability to assign IP address for each system and configuration
2001	Linux VServer is jail mechanism that can partition resources, similar to FreeBSD Jails

2004	First public beta of Solaris Containers released - combines system resource controls and boundary separation provided by zones
2005	Open Virtuozzo is operating system-level virtualization technology for Linux which uses patched Linux kernel for virtualization, isolation, resource management and checkpointing
2006	Process Containers, launched by Google, was designed for limiting, accounting and isolating resource usage of collection of processes - renamed to <i>Control Groups (cgroups)</i> and merged into Linux kernel
2008	Linux Containers (LXC) was the first, most complete implementation of a Linux container manager - implemented using cgroups and Linux namespaces

2011	Warden from Cloud Foundry can isolate environments on any operating system, runs as a daemon and provides API for container management
2013	Let Me Contain That For You (LMCTFY) kicked off as open-source version of Google's container stack - providing Linux application containers
2013	Docker emerged
2016	Singularity (later Apptainer) was released

Docker

What is it? 🔍

- a technology for packaging an application with all of its dependencies into an independent executable unit (a container)
- the container image can be shipped and run consistently across different computing environments

Which benefits does it provide?



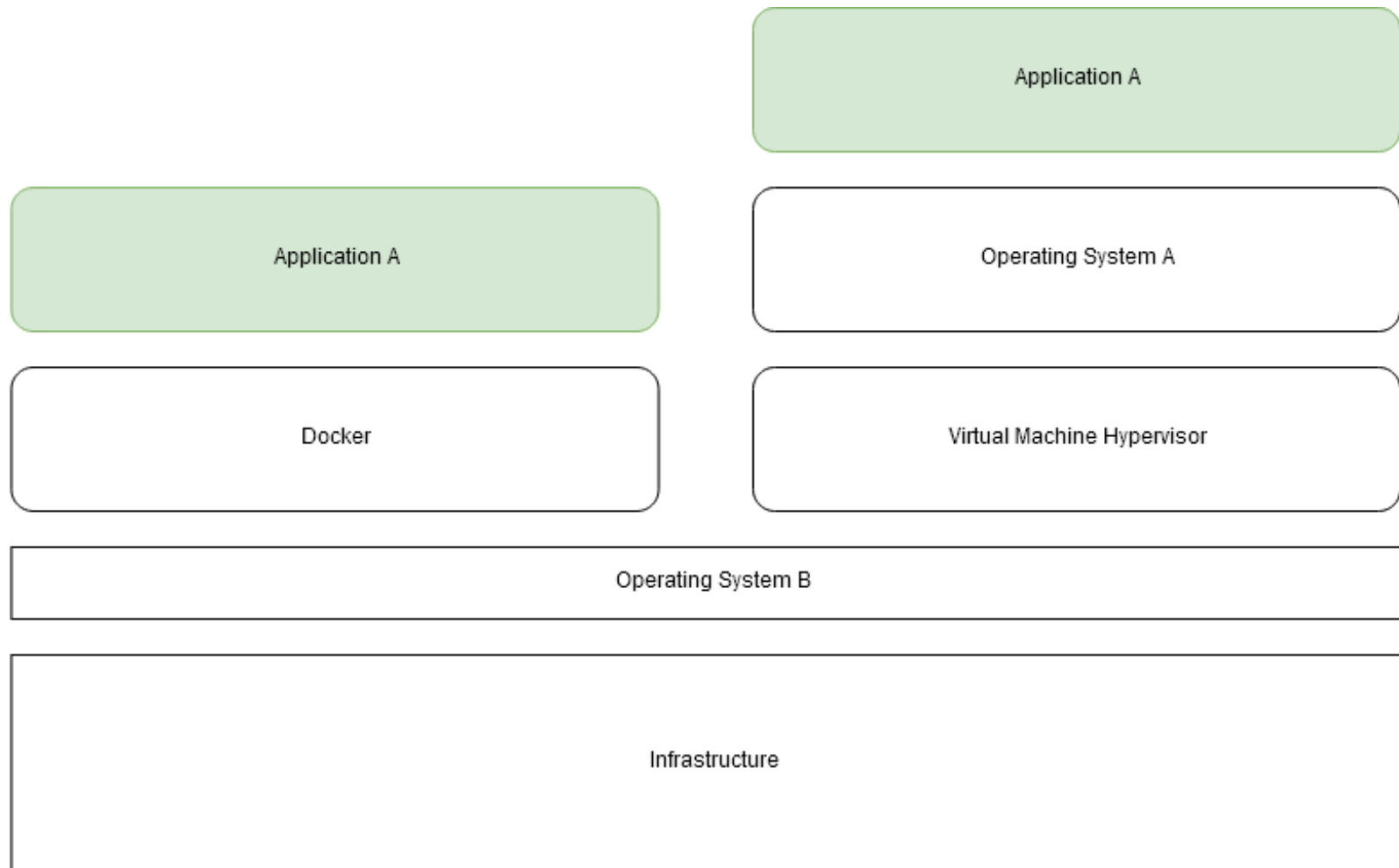
1. Consistency
2. Isolated environments
3. Portability
4. Efficiency

<https://scalablehuman.com/2023/10/24/the-fallacy-of-it-works-on-my-machine>



How is it different from virtual machines?

- containers use and share host OS's kernel, which makes them more lightweight and efficient
- virtual machines emulate entire physical machines, including the complete OS, making it possible to run multiple OS instances on a single physical machine



<https://dev.to/swikritit/docker-for-dummies-introduction-to-docker-5h67>

Key Concepts 🔑

1. **Image:** a standardized package that includes all files, binaries, libraries, and configurations necessary to run a given container
2. **Container:** a running instance of an image, operating in an isolated runtime environment
3. **Dockerfile:** instructions on how to build an image
4. **Docker Hub:** a public registry where developers can share their pre-built images

Key Components 🔑

1. **Docker Engine:** core, open-source technology that builds and runs containers
2. **Docker Client:** command-line tool that sends instructions to the Docker Daemon using REST APIs
3. **Docker Daemon (`dockerd`):** receives and processes API requests and calls container runtime
4. **Container Runtime (`containerd`):** turns static container image into running, isolated application on host OS; industry standard

Let's explore! 🌐

Start by going to https://github.com/InPreD/26-09_bioinfo_ws_docker_and_ci and create a fork.

The screenshot shows the GitHub interface for the repository '26-09_bioinfo_ws_docker_and_ci'. At the top, there are buttons for 'Edit Pins', 'Watch' (0), 'Fork' (0), and 'Star' (0). Below these, a dropdown menu is open, showing the option '+ Create a new fork' highlighted with a red rectangle. The repository is owned by 'marrip' and has 1 branch and 0 tags. The main content area shows a list of files and their commit history:

File	Commit Message	Time
.devcontainer	chore: add devcontainer setup	9 minutes ago
src/greeter	feat: add simple python project	8 minutes ago
.gitignore	chore: add devcontainer setup	9 minutes ago
LICENSE	Initial commit	1 hour ago
README.md	Initial commit	1 hour ago
pyproject.toml	feat: add simple python project	8 minutes ago

Below the file list, there is a section for the README, which includes the title '26-09_bioinfo_ws_docker_and_ci' and the description 'codespace template for docker and ci workshop'. On the right side, there are links to 'workshop', 'Readme', 'AGPL-3.0 license', 'Activity', 'Custom properties', '0 stars', '0 watching', '0 forks', 'Audit log', 'Report repository', 'Releases', and 'Packages'.

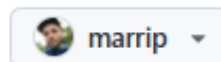
Create your own fork by clicking on **Create fork**.

Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project.

Required fields are marked with an asterisk ().*

Owner *



Repository name *

/ 26-09_bioinfo_ws_docker_

✓ 26-09_bioinfo_ws_docker_and_ci is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

codespace template for docker and ci workshop

45 / 350 characters

☒ Copy the `main` branch only

Contribute back to InPreD/26-09_bioinfo_ws_docker_and_ci by adding your own branch. [Learn more.](#)

You are creating a fork in your personal account.

Create fork

In the forked repository, navigate to **Code** > **Codespaces** > **Create codespace on main**.

The screenshot shows a GitHub repository page for '26-09_bioinfo_ws_docker_and_ci' (Public). The repository is on the 'main' branch, has 1 branch and 0 tags. The file list includes: .devcontainer (chore: add devcontainer), src/greeter (feat: add simple python project), .gitignore (chore: add devcontainer), LICENSE (Initial commit), README.md (Initial commit), and pyproject.toml (feat: add simple python project). The repository is licensed under AGPL-3.0. The 'About' section describes it as a 'codespace template for docker and ci workshop'. The 'Releases' and 'Packages' sections indicate no releases or packages are published.

The 'Code' dropdown menu is open, showing the 'Codespaces' option. A modal window titled 'Codespaces' is displayed, stating 'No codespaces' and 'You don't have any codespaces with this repository checked out'. A red box highlights the 'Create codespace on main' button. Below this button is a link to 'Learn more about codespaces...'. At the bottom of the modal, it says 'Codespace usage for this repository is paid for by marrip.'

Inside the terminal, run the following commands:

```
# check for running docker containers
$ docker ps
# check for existing images
$ docker images
# pull image
$ docker pull ubuntu:26.04
# run image
$ docker run ubuntu:26.04
```

Run an interactive container:

```
# start interactive container
$ docker run -it ubuntu:26.04
# print container os version
@ cat /etc/lsb-release
# exit container
@ exit
# print os version
$ lsb_release -a
```

Instruct docker to remove containers after use:

```
# check for any docker containers
$ docker ps -a
# remove all stopped containers
$ docker rm <CONTAINER ID>
# run docker with --rm flag
$ docker run -it --rm ubuntu:26.04
# exit container
@ exit
# check for all docker containers
$ docker ps -a
```

Remove the docker image:

```
# remove docker image
$ docker rmi ubuntu:26.04
# alternatively check for container image id
$ docker images
# and remove the image by using the id
$ docker rmi <ID>
```

Building an image

We start off by creating a `Dockerfile` in the root directory of our repository (`/workspaces/26-09_bioinfo_ws_docker_and_ci`). We add the following to our file:

```
FROM python:3.14-slim-trixie
RUN echo "Hello world!" > greetings.txt
```

And then we build and run our docker image:

```
# build tagged docker image
$ docker build . -t greeter:test
# run tagged docker image
$ docker run --rm greeter:test cat greetings.txt
```

Let us add a label to our `Dockerfile` to indicate who the author and maintainer is:

```
FROM python:3.14-slim-trixie
LABEL org.opencontainers.image.authors="martin.rippin@helse-bergen.no"
RUN echo "Hello world!" > greetings.txt
```

And then we build and inspect our docker image:

```
# build tagged docker image
$ docker build . -t greeter:test
# run tagged docker image
$ docker inspect greeter:test
```


Next, we are setting `cat greetings.txt` as a default command that is run whenever the container is started:

```
FROM python:3.14-slim-trixie
LABEL org.opencontainers.image.authors="martin.rippin@helse-bergen.no"
RUN echo "Hello world!" > greetings.txt
CMD ["cat", "greetings.txt"]
```

And then we build and run our docker image:

```
# build tagged docker image
$ docker build . -t greeter:test
# run tagged docker image
$ docker run --rm greeter:test
```

There is a small python application in this repository that we would like to include in our docker image:

```
FROM python:3.14-slim-trixie
LABEL org.opencontainers.image.authors="martin.rippin@helse-bergen.no"
WORKDIR /usr/src/greeter
COPY pyproject.toml ./
COPY src/ ./src/
RUN pip install .
CMD ["greeter"]
```

And then we build and run our docker image:

```
# build tagged docker image
$ docker build . -t greeter:test
# run tagged docker image
$ docker run --rm greeter:test
```

A full list of `Dockerfile` instructions can be found here:

<https://docs.docker.com/reference/dockerfile#overview>

Apptainer

What is it?

- container platform designed for ease-of-use on shared systems and in high performance computing (HPC) environments

How is it different from Docker?

Apptainer	Docker
without root-privileges by default	requires root privileges for most operations
SIF (Singularity Image Format) – immutable, portable, cryptographically signed	Docker/OCI images – layered file systems, mutable by default
can run Docker/OCI images	cannot run SIF images

Let's explore! 🌐

Similar to Docker, Apptainer provides a cli:

```
# check for any apptainer container
$ apptainer instance list -a
# pull image
$ apptainer pull docker://ubuntu:26.04
# check identity
$ whoami
# check identity in container
$ apptainer exec docker://ubuntu:26.04 whoami
```

The container image is saved as a `.sif` -file to the working directory.

Building an image 🛠️

We are creating a file called `greeter.def` in the root of our repository and add the following lines:

```
Bootstrap: docker
From: python:3.14-slim-trixie

%post
    echo "Hello world!" > /usr/src/greetings.txt
```

And then we build and run our apptainer image:

```
# build apptainer image
$ apptainer build greeter.sif greeter.def
# execute apptainer image
$ apptainer exec greeter.sif cat /usr/src/greetings.txt
```

A full list of apptainer definition file sections can be found here:

https://apptainer.org/docs/user/main/definition_files.html#sections

3. Continuous Integration (CI)



What is it?

- automating integration of code changes from multiple contributors into single software project
- developers frequently merge code changes into central repository where automated tools are used to assert new code's correctness before integration (test, lint, build)

Which benefits does it provide?

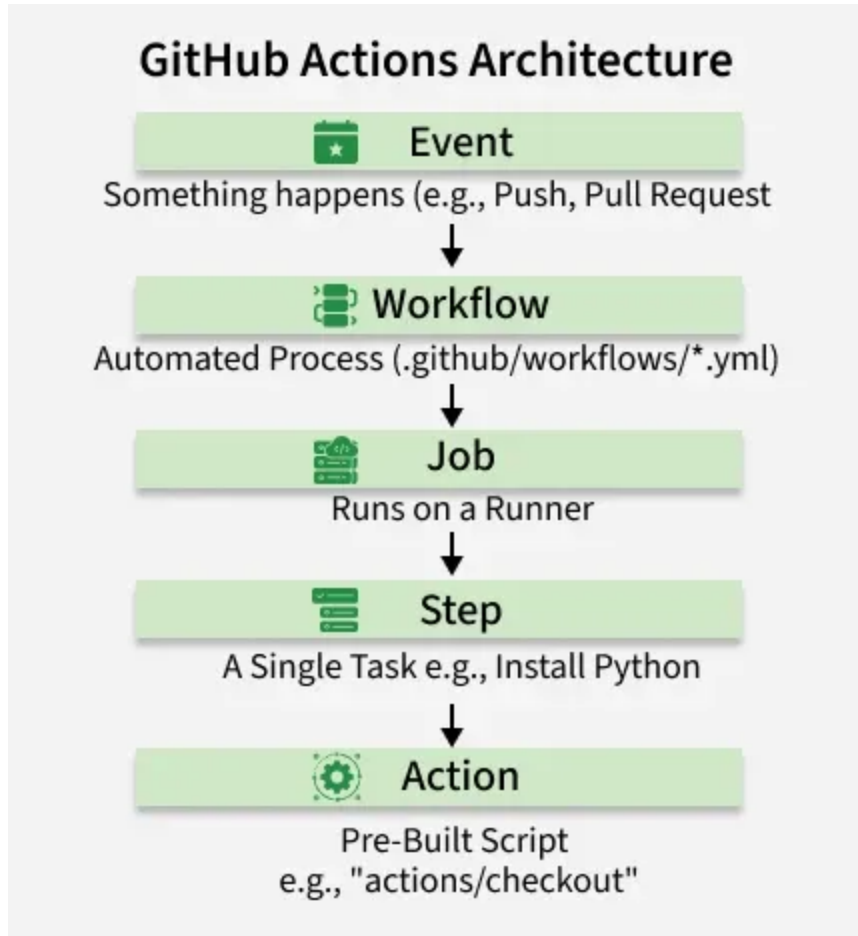
- scale up delivery output
- enables parallel work on features

GitHub actions

What is it? 🔍

- GitHub's continuous integration automation platform
- repetitive tasks are triggered based on code pushes, pull requests, or custom schedules

How does it work? 🤔



<https://www.geeksforgeeks.org/git/introduction-to-github-actions>

Workflow:

- `.yaml` -file located at `.github/workflows/` containing CI instructions

Event

- defined in workflow file
- trigger to start workflow, e.g. push, pull_request, schedule, etc.

Jobs

- defined in workflow file
- contains several tasks/*steps* and is running on a specific runner (server executing the code of the workflow)
- jobs run in parallel or can depend on each other, e.g. testing before building a container image

Step

- single task inside a job, e.g. installing python

GitHub Action

- reusable steps written by other developers that are freely available
- shorter workflows and avoiding unnecessary repetition

Example

```
name: Hello World
# event definition
on: [push]

jobs:
  # job definition
  say-hello:
    # runner definition
    runs-on: ubuntu-latest

    steps:
      # github action definition
      - name: Checkout Code
        uses: actions/checkout@v4

      # step definition
      - name: Run a Hello World Script
        run: |
          echo "Hello, world!"
          echo "The current repository branch is ${ github.ref }"
```


Let's explore! 

We are adding a workflow file to our repository:

```
# create .github/workflows  
$ mkdir -p .github/workflows
```

Now we add `.github/workflows/hello_world.yaml` using the example file from the previous slide.